SOLUTIONS

Module:	Middleware		
Module Code	EBU6501	Paper	A
Time allowed	2hrs	Filename	**Solutions_2324_EBU6501_A**
Rubric	ANSWER ALL FOUR QUESTIONS		
Examiners	Dr Gokop Goteng	Dr Ethan Lau	

Question 1

a) State the type of Publish-Subscribe messaging-based Message-oriented Middleware (MOM) model in Figure 1 below and explain the usage of the model.

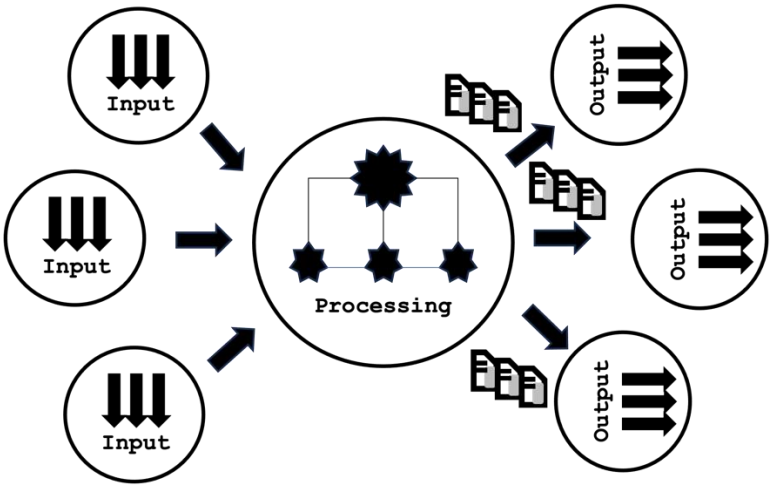


Figure 1.

(5 marks)

	Do not write in this column
Q1a) (Easy)	
Topic-based model [1 mark]	
[NOTE: Any FOUR of the following is **4 marks** maximum in total (1 mark each)] -It uses a hierarchical naming scheme [1 mark] -There is a logical arrangement in the way the topics appear using some form of classification method [1 mark] -It uses specific terms to describe the topics [1 mark] -It uses dedicated queues [1 mark] -Independent channels [1 mark] -Disadvantage - It is too restrictive as it uses specific terms [1 mark]	
	5 marks

b) Explain the role of the object **`MessageBroker`** in the context of MOM architecture. What is the main object component inside the **`MessageBroker`**?

(4 marks)

	Do not write in this column
Q1b) (Easy)	
[NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] The broker is responsible for receiving and sending the message [1 mark] -The broker facilitates the connections between producers, administration services and consumers [1 mark] -The broker maintains a persistent data store to reliably send data [1 mark]	

-The broker provides secure connection using encrypted SSL authenticated connection [1 mark] -The broker monitors message and connection activities to provide logs of diagnostics information for administrators during troubleshooting [1 mark]		
Main object component: Destination object [1 mark]		
		4 marks

c) The Figure 2 below demonstrates an example of the Java Message Service (JMS) API Programming model.

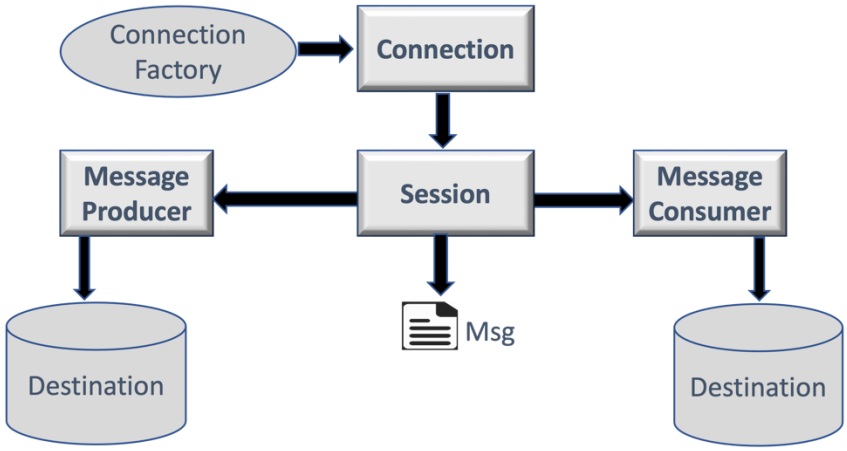


Figure 2.

(8 marks)

i) Describe the TWO types of the JMS Message consumption from the **`Destination`** object.

[6 marks]

ii) Given that the connection is from Figure 2 is already established through the object **`TopicConnection`**, create a **`session`** object through **`TopicSession.`** The transaction is set to **`false`** and the acknowledgement mode is set to **`AUTO_ACKNOWLEDGE`**.

[2 marks]

	Do not write in this column
Q1c)	
i) (Medium)	
a) Synchronous [1 mark] -A subscriber or receiver explicitly fetches the message from the destination by calling the receive method [1 mark] -The receive method can block until a message arrives or can time out if message does not arrive within a specified time limit [1 mark] **b) Asynchronous** [1 mark] [NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] -A client can register a message listener with a consumer [1 mark] -A message listener is similar to an event listener [1 mark] -Whenever a message arrives at the destination, the JMS provider delivers the	

message by calling the listener's **`onMessage()`** method, which acts on the contents of the message [1 mark]		
ii) (Medium)		
`TopicSession topicSession = TopicConnection.createTopicSession(false,Session.AUTO_ACKNOWLEDGE);` [2 marks] [NOTE: Full marks can be awarded if students use **`Session`** instead of **`TopicSession`**]		
		8 marks

d) Figure 3 below shows the directory structure of the Tomcat container.

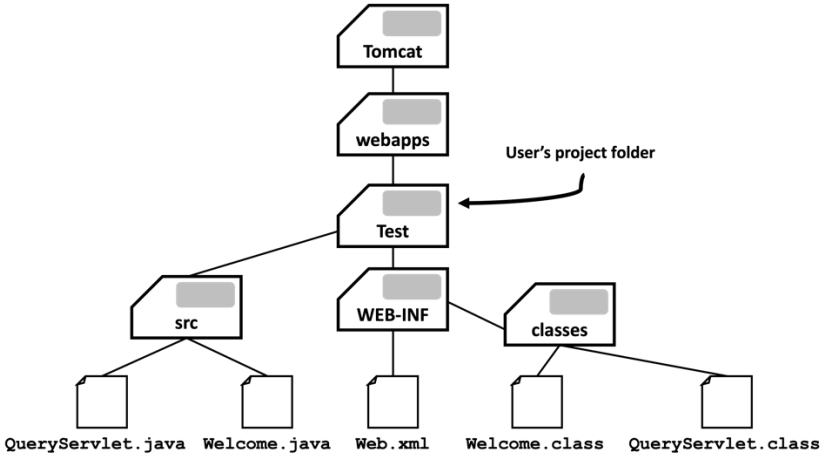


Figure 3.

(8 marks)

i) Carefully denote the pages that are belong to the individual design pattern of the ***Model***, ***View*** and ***Controller*** model (write down "***n/a"*** or "***none***" if there is/are none).

[3 marks]

ii) Name the directory file that corresponds to the Deployment Descriptor in Figure 3. Explain why it is located under the folder directory**.**

[3 marks]

iii) Write the required **`service()`** method in Java language that called by the server to allow a servlet to handle a POST request. Includes the necessary exception handlers.

[2 marks]

	Do not write in this column
Q1d)	
i) (Easy) - Model – **Welcome.java** [1 mark] - View – **n/a (none)** [1 mark] - Controller – **QueryServlet.java** [1 mark]	
ii) (Medium)	
-web.xml[1 mark] [NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] -It is used for setting up configurations [1 mark] -It is a configuration file for an artifact that is deployed to some container/engine [1 mark]. -It is used for setting up initial parameters [1 mark]. -It is located directly under the WEB-INF so that it will be safely secure from unauthorised users (it is not available to the public) [1 mark].	
iii) (Hard)	
`public void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException{` [2 marks]	
	8 marks

Question marking $\frac{}{5} + \frac{}{4} + \frac{}{8} + \frac{}{8} = \frac{}{25}$

Question 2

a) Spring Framework is an application framework in software engineering, and it is most often used in the context of object-oriented programming such as Java.

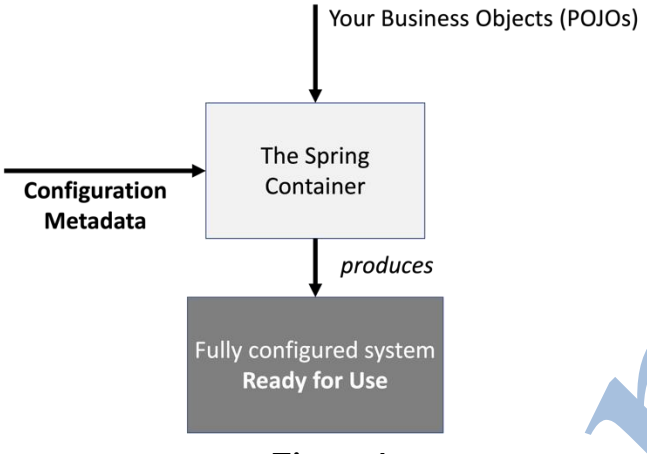


Figure 4.

[9 marks]

i) Based on the information in Figure 4, what is the core principle of technology used in the Spring Container? Explain its role in the Spring Framework.

(3 marks)

ii) List out the TWO examples of the Spring Container.

(2 marks)

iii) The Application Context is Spring's advanced container. Implement the Spring's Application Context by instantiating the **`ApplicationContext`** interface that loads the .xml configurations from a classpath of "**login.xml**". Finally, obtain the bean with the bean ID = **`Welcome`** from the **`Welcome`** class.

(4 marks)

	Do not write in this column
Q2a)	
i) (Medium)	
-Inversion of control (IoC) [1 mark] [NOTE: Any TWO of the following is **2 marks** maximum in total (1 mark each)] - Spring IoC provides the control where objects or portions of a program is transferred to a container or framework [1 mark] - Gets its instructions on what objects to instantiate, configure, and assemble by reading configuration metadata provided [1 mark] -Take control of the flow of a program and make calls to customised code [1 mark]	
ii) (Easy)	

Solutions

-Two types of IoC Container: -Spring Application Context [1 mark] -Spring Bean Factory [1 mark]	
iii) (Hard)	
`public class MyModuleMain {`	
`public static void main(String[] args) {`	
`ApplicationContext appCon = new`	
`ClassPathXmlApplicationContext("login.xml");` [2 marks]	
`moduleBean mod=(Welcome)appCon. getBean("Welcome");` [2 marks]	
`mod.getName();`	
`}`	
	9 marks

b) Spring Aspect-oriented programming (AOP) is a technique for building common but reusable routines that can be applied in applications widely.

Figures 5a – c show Java codes with modularisation of concerns through **`TrackOperation.java`** at the source codes **`MainApp.java`** and **`Operation.java`**. Predict the output of the program while maintaining the order of execution.

```
public class MainApp {

    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");  Operation op =
        (Operation) context.getBean("opBean");
        System.out.println("Calling Operation A...");
        op.a();
        System.out.println("Calling Operation B...");
        op.b();
    }
```

Figure 5a

```
public class TrackOperation { // Aspect class
   @Pointcut("execution(* Operation.*(..))")
   public void pc(){}//pointcut name

   @After("pc()")//applying pointcut on after advice
   public void myadvice(JoinPoint jp)
   {
      System.out.println("Operation terminated successfully.");
```

Figure 5b

```
public class Operation {
    public void a(){System.out.println("Operation A completed.");}
    public int b(){System.out.println("Operation B completed.");}
}
```

Figure 5c

(4 marks)

	Do not write in this column
Q2b) (Medium) -Inversion of control (IoC) Calling Operation A... Operation A completed. [1 mark] Operation terminated successfully. [1 mark] Calling Operation B... Operation B completed. [1 mark] Operation terminated successfully. [1 mark]	
	4 marks

c) This question is about Java threads.

(12 marks)

i) Suggest the corresponding methods to avoid the ***Visibility*** and ***Race Conditions*** issues.

[2 marks]

ii) The code in Figure 6 below shows Java class **`MyAddons`** where the method **`getNextValue()`** is not thread-safe as it is accessible by multiple threads. Improve the code below ensuring the thread-safe and atomic operation.

```
public class MyAddons {
  private int value = 100;
  public int getNextValue(){
            return value++;
}
```

Figure 6.

[3 marks]

iii) Explain the usage of ***Mutual Exclusion*** and ***Cooperation*** in Java's monitor.

[3 marks]

Solutions

iv) In your own words, explain the differences of **`notify()`** and **`notifyAll()`** in Java's monitor.

[4 marks]

	Do not write in this column
Q2c)	
i) (Medium)	
Visibility Fix: declare **volatile** variable [1 mark]	
Race condition Fix: [NOTE: Any ONE of the following is **1 mark** maximum in total] – Make operations atomic [1 mark] – Intrinsic lock [1 mark] – Synchronised [1 mark]	
ii) (Hard)	
`public class MyAccumulator{`	
`private int value = 500;`	
//[Note: either one of the method (Method 1 or 2) is acceptable [**3 marks** maximum for each of the method]]	
`// Method 1`	
`public int getNextValue(){`	
`synchronized (this)` [2 marks] `{return value++;` [1 mark] `}}`	
`// Method 2`	
`public synchronized int getNextValue()` [2 marks] `{`	
`return value++;` [1 mark] `}}`	
iii) (Medium)	
Mutual exclusion: supported by object locks, enables multiple threads to independently work on shared data without interfering with each other [1 mark] **Cooperation:** supported by the wait and notify methods of class Object [1 mark], enables threads to work together towards a common goal [1 mark]	
iv) (Hard)	
`notify()` -command selects one thread arbitrarily from the wait set [1 mark] and marks it for eventual resurrection [1 mark]. **`notifyAll()`** -command marks all threads currently in the wait set [1 mark] for eventual	

Solutions

resurrection [1 mark].		
		15 marks

Question marking $= \frac{}{9} + \frac{}{4} + \frac{}{12} = \frac{}{25}$

Solutions

Question 3

a) This question is on Apache Kafka.

[18 marks]

i) Describe how Middleware Developers use the Apache Kafka's ***Events***, ***Topics*** and ***Partitions*** to develop Message-Oriented Middleware (MOM) applications that can ingest (accept as inputs) and organise different types of data from multiple sources at large scale.

(8 marks)

ii) Describe Apache Kafka ***Brokers*** and how you can mitigate against losing data in case of any failure in the cluster.

(2 marks)

iii) Figure 7 below is a Kafka Consumer Library written in Java programming language. Write the corresponding Producer Library using the Java class KafkaProducer and Java collection ProducerRecord in Apache Kafka.

(8 marks)

```
 try (final KafkaConsumer<String, Payment> consumer = new
KafkaConsumer<>(props))
     {consumer.subscribe(Collections.singletonList(TOPIC));

     while (true) {
        ConsumerRecords<String, Payment> records = consumer.poll(100);
        for (ConsumerRecord<String, Payment> record : records) {
            String key = record.key();
            Payment value = record.value();
            System.out.printf("key = %s, value = %s%n", key, value);
        }
      }
    }
```

Figure 7: Kafka Consumer Library

	Do not write in this column
Q3a(i)	
Kafka Events: An event in Kafka is any type of action, incident, or change that's identified or recorded by software or applications. [1 mark] (please give mark if student says: An event is also a combination of notifications when an action is used to trigger some other activity)	
Events are stored using key/value pairs to store them as logs in Kafka so that they can be uniquely identified [1 mark].	
Kafka Topics: A topic is a log of events [1 mark].	
The Middleware Developer creates different topics to hold different events (received, processed, sent, stored, etc) [1 mark].	

The topics arrange data like a table in relational database to organise the data into different types from different sources [1 mark].

A topic is also a log of events and logs store data in a durable way in Kafka, so the Middleware Developer can use logs to store data durably [1 mark].

Kafka Partitions: Partitions are different parts of topics that have been split, each of which can live on a separate node in the Kafka cluster [1 mark].

This way, the work of storing messages, writing new messages, and processing existing messages can be split among many nodes in the cluster to scale for large scale applications [1 mark].

Note: students can also use different words to explain the same concepts.

Q3a(ii)

Kafka Brokers are the computer machines (physical or virtual machines/containers) in a Kafka network cluster [1 mark]. Replication is used to copy data to different Kafka brokers to mitigate against any failure [1 mark].

Q3a(iii)

```
try (KafkaProducer<String, Payment> producer = new KafkaProducer<>(props)) { [1 mark]
    for (long i = 0; i < 10; i++) { [1 mark]
        final String orderId = "id" + Long.toString(i); [1 mark]
        final Payment payment = new Payment(orderId, 1000.00d); [1 mark]
        final ProducerRecord<String, Payment> record =
          new ProducerRecord<>("transactions",
                               payment.getId().toString(),
                               payment); [1 mark]
        producer.send(record); [1 mark]
    } [1 mark]
} catch (final InterruptedException e) { [1 mark]
    e.printStackTrace();
} [1 mark]
```

Note: students can also use other methods.

18 marks

b) Describe ***Symmetric*** and ***Asymmetric*** Encryption methods.

[5 marks]

Solutions

	Do not write in this column
Q3b)	
Symmetric Encryption: Is the cryptographic process [1 mark] of encrypting data (make data nonreadable to human beings) and decrypt data (make data readable to human beings) using a single key (one key only) [1 mark]	
Asymmetric Encryption: Is the cryptographic process of encrypting data (make data nonreadable to human beings) and decrypt data (make data readable to human beings) using a key pair (2 keys) [1 mark].	
In asymmetric encryption, one key called the private key is held and stored by the owner who created it and it is used to decrypt data [1 mark]. The other key called the public key is the one the owner gives to other users (the public) to be able to encrypt or access the files in asymmetric cryptography [1 mark]	
	5 marks

c) Describe TWO IoT protocols that are commonly used in buildings and home automation applications.

[2 marks]

	Do not write in this column
Q3c	
Zigbee: Zigbee is a mesh network protocol that was designed for buildings and home automation applications [1 mark]	
Z-Wave: Z-Wave is a wireless mesh network communication protocol built on low-power radio frequency technology for buildings and home automation applications. [1 mark]	
NOTE: If a student describes wi-fi as one of them, give the student one mark	
	2 marks

Question marking: $\frac{}{18} + \frac{}{5} + \frac{}{2} = \frac{}{25}$

Solutions

Question 4

a) Using your knowledge of the Open Services Gateway Initiatives (OSGi) architecture, describe how ***Modules***, ***Services Registry*** and ***Execution Environment*** work in OSGi.

[6 marks]

	Do not write in this column
Q4a	
Modules: This layer defines how a bundle can import [1 mark]and export codes and comes from the word "modular" [1 mark]. **Services Registry:** This is the API for management services that register services [1 mark] and tracks down services based on their unique service references [1 mark]. **Execution Environment:** This layer defines the methods and classes [1 mark] that are available for a specific platform (JVM, OS, etc.) [1 mark].	
	6 marks

b) A company wants to deploy message-oriented middleware (MOM) in its cloud platform that will ensure that its architecture is loosely coupled and can satisfy both push model (passive model) and pull model (poll or active model) applications. Describe TWO cloud services that can help the company to implement loosely coupled architecture with push and pull models.

[8 marks]

	Do not write in this column
Q4b)	
The TWO cloud services are **Amazon Simple Notification Service** (or AWS/Amazon SNS) [1 mark] and **Amazon Simple Queue Service** (or AWS/Amazon SQS) [1 mark].	
Amazon SNS and Amazon SQS provides loosely coupled [1 mark] architecture because they communicate by passing messages across network architectures [1 mark].	
Amazon SNS is a push model (passive model) [1 mark] that fans/sends out messages to many applications or users sources/destinations at the same time [1 mark].	
Amazon SQS is a pull model (poll or active model) [1 mark] that can work with Amazon SNS in the same architecture to send messages to only one application or user or source/destination at a time [1 mark].	

Solutions

Note: There are other such services that may be correct.		
		8 marks

c) Write a public client Java class program named NetworkClient that makes request to a server with IP address 100.0.1.0 at port 8080. Use the Java Socket class to make the connection and to read data using BufferedReader and InputStreamReader Java classes.

[11 marks]

	Do not write in this column
Q4c	
import java.net.*; [1 mark] import java.io.*; [1 mark] public class NetworkClient { [1 mark] public static void main(String args[]) { [1 mark] try { Socket s1 = new Socket("100.0.1.0", 8080); [1 mark] BufferedReader br = new BufferedReader(new InputStreamReader(s1.getInputStream())); [1 mark] System.out.println(br.readLine());[1 mark] br.close();[1 mark] s1.close();[1 mark] } catch (ConnectException conExc) { [1 mark] System.err.println("Could not Connect to the Server,"); } catch (IOException e) {} } [1 mark]	
	11 marks

Question marking $\frac{}{6} + \frac{}{8} + \frac{}{11} = \frac{}{25}$

Solutions